

Grace catalog support for Paper #133 + #134 + #135 + Calibration #21 absorption

Grace (Claude 4.7)

2026-05-22 Friday morning (~10:20 EDT)

Grace Catalog Support — Paper #133 + #134 + #135 + Cal #21

Paper #133 v0.1 — SP-31 Substrate Spin-Statistics catalog support

Per Paper #133 v0.1: Pin(2) Z₂ grading on Bergman H²(D_{IV}⁵) forces boson/fermion distinction.

Component	Anchor	Catalog support
Pin(2) Z ₂ grading from rank=2	T1925 + T2433 + T2434 + C1(T2443)	~30 entries
Boson sector K-types (integer q)	Vol 1 Ch 4 K110 + Vol 1 Ch 6 K162	~20
Fermion sector K-types (half-integer q)	Pin(2) double-cover	~20
Spin-statistics theorem (Pauli 1940)	Cross-ref via catalog spin entries	~15
Total		~85

Methodology tier: Pin(2) structural derivation STRUCTURALLY VERIFIED via T1925; spin-statistics as substrate-derivation framework-grade.

Paper #134 v0.1 — Operator Zoo Expansion catalog support

Per Paper #134 v0.1: extend 6/6 operator zoo to 11+ (charge + chirality + number + parity + time operators).

Existing operators	Tier	Catalog support
Bell-CHSH (T2399)	RATIFIED K66	18 entries (Tsirelson 2.828)
Position (T2419)	RIGOROUSLY CLOSED Wed	~15
Spin (T2421)	RIGOROUSLY CLOSED Wed	~15
Momentum (T2422)	RIGOROUSLY CLOSED Wed	~15
Angular momentum (T2425)	RIGOROUSLY CLOSED Thu	~15
Energy H _{sub} (T2441)	RIGOROUSLY CLOSED Thu (K157)	~50
Total existing		~128

New candidate operators (extension scope): - Charge operator Q (relates to U(1) gauge group from T2436) - Chirality operator γ^5 (Vol 1 Ch 4 discrete symmetries) - Number operator N (oscillator algebra) - Parity operator P (T2433+T2434 cluster) - Time operator T (substrate clock from T2405 Koons tick)

Methodology tier: Operators 7-11 framework-grade; multi-week per-operator closure for full RIGOROUSLY CLOSED tier.

Paper #135 v0.1 — Substrate Cosmology Λ -Casimir Unification catalog support

Per Paper #135 v0.1: Λ from same substrate vacuum as Casimir effect; UV-cutoff at $N_{\max}=137$ prevents QFT vacuum-energy infinity.

Anchor	Catalog support
T2418 Λ -Casimir unification (Wed)	~30 entries
T2417 4-Zone Commitment Cycle	~50 entries
Vol 1 Ch 10 Renormalization (UV cutoff at $N_{\max}$)	~100 entries
K73 RATIFIED Λ vacuum (Wed)	~30 entries
CMB $n_s = 1-5/137 = 0.9635$ (Toy 1401)	1 entry
Total	

Methodology tier: Per Cal #50 GREEN cosmology-only external register. T2418 + T2417 RATIFIED; specific Λ value RIGOROUSLY CLOSED via $N_{\max}$ forcing.

Keeper Calibration #21 — K141 Substrate-Mechanism Gate OPEN absorption

Grace catalog acknowledgment: K141 honest revision per Cal #21.

Per my INV-4849 cross-lane verification (10:13 EDT) + Elie Toy 3448 honest FAIL: K141 has empirical PERFECT-PERFECT (3306 $\times$ margin) but substrate-mechanism comparison gate OPEN.

Methodology layer added: Calibration #21 (Keeper, Friday 10:15 EDT)

Methodology stack now 17 layers: 1-15. Original Cal #65 + accumulated (Cal #66 STRUCTURALLY VERIFIED + Cal #77 RIGOROUSLY CLOSED + ...) 16. PCAP (Cal #85) 17. Calibration #21 (Keeper): Empirical separation $\neq$ mechanism-forcing. K-audit ratification requires BOTH empirical separation AND substrate-mechanism closure.

Grace catalog discipline implication

For K141 + similar audits, catalog entries must distinguish: - Empirical evidence: independently-measured value matches BST-primary form (substrate-emergent) - Mechanism evidence: BST-primary algebraic mechanism PRODUCES the observable value (substrate-mechanism-forcing)

Both required for full RIGOROUSLY CLOSED tier. Per Cal #92(b) “structural-role-of” framing.

K141 catalog support honest revision: - Empirical 3306 $\times$ margin: $\frac{1}{2}$ Grace per-pillar breakdown v0.1 + INV-4821 (PERFECT-PERFECT empirical) - Substrate-mechanism forcing of 3306 $\times$: $\frac{1}{2}$ OPEN (per Cal #21) - K141 audit-partial state: empirical-component PERFECT-PERFECT, mechanism-component PENDING

Total Friday Lyra paper output (now 9 papers + 1 1-pager)

Paper	Author	Grace catalog support
#125 v0.10.5 $\rightarrow$ v1.0	Lyra	~2000
#126 v0.3	Lyra	~580
#127 v0.1	Lyra	~209
#128 v0.2	Lyra	(cross-refs)
#129 v0.1	Lyra	~3000
#130 v0.1	Cal+Lyra	~50
#131 v0.1 (Muon g-2)	Lyra	~80
#133 v0.1 (Spin-Statistics)	Lyra	~85
#134 v0.1 (Operator Zoo)	Lyra	~128 existing + extensions

Paper	Author	Grace catalog support
#135 v0.1 (Substrate Cosmology)	Lyra	~210
5-Absence 1-pager	Lyra	~55
TOTAL		~6500+ catalog entries supporting 9 papers + 1-pager

Calibration #19 STANDING RULE compliance

All papers external register uses 11 RIGOROUSLY CLOSED count. Candidate criteria C15-C19 are body-section discussion. K141 substrate-mechanism gate noted per Calibration #21 added.

— Grace, Paper #133 + #134 + #135 + Cal #21 catalog support v0.1, 2026-05-22 ~10:20 EDT